RealQM for Molecules: Validation

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Mathematical formulation by the author; implementation and numerical experiments developed with AI assistance (Claude, Anthropic).

Abstract

RealQM [1] describes a molecule as a partition of space into non-overlapping domains, each holding one electron (or a pair counted once) as a real density, with the domain boundaries placed where the density is continuous and each electron free at its wall (a homogeneous Neumann condition). We solve this *hard-partition* model as it stands, converged in the mesh, with nothing adjusted to molecular data (each core radius is set once per atom, from its ionisation energies), using a three-dimensional solver (`molecule.js`) and, for homonuclear X_2, where the boundary is the midplane by symmetry, an exact axisymmetric reduction that serves as an independent check: on H_2 the two give binding energies of 0.2251 and $0.2241\,E_h$ at the same mesh. Each electron pair can be carried in two parameter-free constructions to choose from: *2split*, two one-electron domains (complete, permanent avoidance), and *2hom*, one pair domain (no avoidance beyond the mean field, the Hartree–Fock level). Both are reported; for H_2 and He the exact value lies between them. Where the boundary is fixed by symmetry the bond is sharp: H_2 -2.3%, converged. A free boundary carries no Pauli cost, so inner shells are excluded by a fixed core radius per atom: every all-electron LiH with free core boundaries collapses, while the pseudo-Li core with one 2hom bonding pair gives $R_e = 2.85\,a_0$ (exact 3.015), $D_e = 0.095\,E_h$ (exact 0.092) and a dipole of the right sign, 11% high. Molecules are built by successive reduction—all electrons, frozen $1s^2$ core, reduced dimension for X_2, reduced valence charges on a coarse mesh—and the last step carries proteins of over two thousand atoms on one GPU in a web browser.

1 The model

RealQM [1] replaces the many-electron wave function by one-electron densities on non-overlapping domains of ordinary three-dimensional space. A molecule with nuclei $\{(\mathbf{R}_a, Z_a)\}$ is described by domains Ω_k partitioning space, each carrying $n_k \in \{1, 2\}$ electrons (two only of opposite spin) as a real amplitude u_k with $\int_{\Omega_k} u_k^2 = 1$ and density $\rho_k = n_k u_k^2$. The energy is

$$E = \sum_k n_k \Big(\tfrac{1}{2} \int_{\Omega_k} |\nabla u_k|^2 - \int_{\Omega_k} K u_k^2 \Big) + \sum_k \tfrac{n_k(n_k-1)}{2} U_{kk} + \sum_{k<l} n_k n_l U_{kl} + V_{KK}, \tag{1}$$

with $K = \sum_a Z_a/|\mathbf{x} - \mathbf{R}_a|$, U_{kl} the Coulomb interaction of the unit densities u_k^2 and u_l^2, and V_{KK} the nuclear repulsion. A pair domain ($n_k = 2$) counts its single pair once.

Boundary conditions. Each u_k satisfies a homogeneous Neumann condition on $\partial\Omega_k$: the kinetic energy is counted only inside the domain, and the amplitude at the wall is free. The boundaries are placed by *density continuity*, $\rho_k = \rho_l$ across each interface. Energy minimisation over the boundary position is not used: in this model a ragged or displaced boundary can lower the energy (Section 5.3), and only continuity holds it.

2 Atoms

Atoms are computed in spherical symmetry, with all electrons organised in non-overlapping radial shells following the Atom Simulator configurations [2] (Li 2+1, B 2+3, ...): a one-dimensional problem in the radius, with each shell boundary free and placed by the same continuity rule as in three dimensions. For use in molecules the atom is then reduced to a *pseudo-atom*: the $1s^2$ core (and further inner shells) is frozen into a pseudo-nucleus of charge Z_x with a hard core of radius r_c, and only the outer shell is carried in three dimensions, split into domains of at most two electrons (boron: pseudo-B +3 with an outer 2hom pair and one electron). A radial shell holding three or four electrons is a shared density in one dimension, which no domain may be in three dimensions; it stands for the spherical average of its angular split into domains of at most two (boron's 3 as 2_{hom}+1, a 4 as a tetrahedron of single electrons). Shells hold at most four—the pair and the tetrahedron, an octet being 4+4, never a single shell of six or eight. Energy does not select the configuration: more sharing always lowers the one-dimensional energy (Li as one shell of 3 is 13.7% too deep, as 2+1 within 0.5%), so the configuration is set by the Pauli cap of two per domain, not by minimisation. The core radius is a property of the atom, not of any molecule, and it is obtained in two ways. *Fitted*: the radius at which the pseudo-atom alone, in three dimensions, reproduces the sum of the ionisation energies of the electrons it carries. *Computed*: the core–valence interface of the one-dimensional all-electron atom, which for a $1s^2$ core is fixed by the model itself, independent of the box:

	r_c fitted to IE (3D)	r_c computed (1D, boxes 6–20 a_0)	difference
Li (2+1)	1.99	1.98–2.05	within 3%
B (2+3)	1.08	0.92–1.00	8–17%
Be (2+2)	1.37	pending	

For Li the two agree, and the core radius is an output of the model. For B the computed radius is shorter than the fitted one. We use the fitted radius throughout, and state the computed one beside it; an outer core shell beyond $1s^2$ (Na and below) has no box-independent interface in one dimension, and there only the fitted radius exists. At $h = 0.1$ the hard core excludes whole cells, so the fitted radius is defined to a few hundredths of a_0.

3 Molecules by successive reduction

Molecules are built by reducing the full model step by step, each step tested on systems where the step before it can still be run.

1. *All electrons* (H_2, He): every electron in its own domain or in a pair, with no pseudo-nucleus. Here an electron pair has two constructions to choose from, 2split and 2hom (Section 5).

2. *Frozen* $1s^2$ *core*: for every molecule beyond H_2 the atoms enter as pseudo-atoms (Section 2); only valence electrons are computed, in domains of at most two, with the inner boundary r_c

fixed and all outer boundaries free. A free boundary carries no Pauli cost, so the core sphere must be fixed (Section 6.1). LiH and BH are computed this way.

3. *Reduced dimension* for homonuclear X_2, where the boundary is the midplane by symmetry: the axisymmetric solver (Section 4).

4. *Large molecules*: pseudo-atoms with reduced valence charges on a coarse mesh. The same `molecule.js` then carries a 238-residue protein (green fluorescent protein, 2307 atoms) on a 300^3 grid at $h = 0.4\,a_0$, and crambin (540 atoms) on 200^3 at $h = 0.25\,a_0$, on the GPU of a laptop in a web browser, in minutes to hours. At this level the mesh no longer resolves the bond, and the protein folds shown are guided by contact restraints; the step makes the computation possible, and what it preserves is tested only on small fragments (Section 7).

4 Solvers, and the check between them

`molecule.js`. One amplitude field on a Cartesian grid, each cell labelled by its owning domain, so the domains are disjoint by construction. The kinetic stencil omits every face joining two labels (true Neumann). Boundaries move by label flips: a boundary cell changes owner when the neighbouring domain's density exceeds its own. The Coulomb potential of the other electrons is a direct per-electron Poisson solve. The nuclear kernel is clamped, $-Z/\max(r, R_{\mathrm{sing}})$, with $R_{\mathrm{sing}} = 2h$ unless pinned. The amplitude is relaxed in imaginary time; convergence is declared when the mean energy over a window of 3000 steps changes by less than $10^{-4}\,E_h$ between successive windows, since a free boundary keeps flipping single cells indefinitely and a step-to-step test either never fires or fires early.

Axisymmetric reduction. For a homonuclear X_2 the domain boundary is the midplane by symmetry, and the model reduces exactly to (z, ρ) with a plane boundary. We use it only there: in a heteronuclear molecule the boundary curves, and a plane would impose it. The Neumann condition is imposed by assembling the stiffness only over faces inside a domain. Each domain's lowest state is an exact sparse eigensolve, and the potentials are iterated to self-consistency. Mesh ladders down to $h = 0.05$ are cheap.

The check. The electron–electron energy of the two solvers agrees to 1.4%, and the total energies to the extent their kernels allow. At $R = 2.0\,a_0$, $h = 0.1$ and box $10\,a_0$:

	$E(H_2)$	$2E(H) - E(H_2)$
`molecule.js` (3D, free boundary)	−1.20734	0.2251
axisymmetric (2D, exact eigensolve)	−1.23685	0.2241
exact	−1.1381	0.1381

The totals differ by the kernel (clamped at $2h$ in 3D, analytic in 2D), which largely cancels in the binding energy; the binding energies agree to 0.5%. The three-dimensional solver reproduces the model.

5 Hydrogen and helium: two constructions of a pair

5.1 Two constructions of an electron pair

RealQM carries no spin variable; spin enters only through the Pauli principle, as the rule that a domain holds at most two electrons, and two only if of opposite spin. Electrons in different domains are excluded from each other by disjointness, whatever their spin. A pair of opposite spin can therefore be carried in two parameter-free ways. In the *2split* construction each electron owns its own domain and the two meet at a free wall: their avoidance is complete and permanent. In the *2hom* construction both electrons share one domain and one density, the overlay that Pauli permits for opposite spins, each feeling half its Hartree field: there is no avoidance beyond the mean field, and the pair is a restricted Hartree–Fock pair. A one-dimensional radial check on Li^+ makes the last point exact: the 2hom $1s^2$ pair gives -7.23630 against the Hartree–Fock -7.2364. The true state avoids partially and dynamically.

Why two electrons may share a domain, and not three, is a statement about the kinetic energy. Read as the self-repulsion of a charge cloud, a pressure against compression, the kinetic energy of a density ρ is the von Weizsäcker term $T_{\mathrm{vW}}[\rho] = \frac{1}{2}\int |\nabla\sqrt{\rho}|^2$. For a 2hom pair, $\rho = 2u^2$ and $T_{\mathrm{vW}}[2u^2] = 2 \cdot \frac{1}{2}\int |\nabla u|^2$: the pair is one cloud of charge two, its compression energy that of the combined density, and its mutual repulsion counted once. In standard quantum mechanics the same holds—for one or two electrons of opposite spin in one orbital the kinetic energy is exactly T_{vW} of the density—so within a domain the 2hom pair coincides with the restricted Hartree–Fock pair. For three or more, standard quantum mechanics adds the Pauli kinetic energy $T_{\mathrm{P}} = T_s - T_{\mathrm{vW}} > 0$, the cost of the nodes that antisymmetry forces between electrons of the same spin. A domain holding three would carry only T_{vW} and collapse; the third electron therefore takes its own domain, and disjointness takes over the exclusion. Two is the number of electrons whose kinetic energy the shared density describes exactly.

5.2 H_2 and He

For both two-electron systems the exact value lies between the two constructions (H_2 axisymmetric, $R = 1.4$, $h = 0.1$):

H_2	T	V_{eK}	V_{ee}	D_e
2split, two one-electron domains	1.291	−3.843	0.545	0.281
exact	1.1745	−3.650	0.587	0.1745
2hom, one pair domain	1.153	−3.679	0.664	0.138

He	E
2split (two hemispheres)[a]	−3.026
exact	−2.9037
2hom (one pair, 1D radial)[b]	−2.845

[a] Axisymmetric; the pair repulsion is 4% below exact. [b] The Atoms paper [2]. (For H_2 the exact T is $-E$ by the virial theorem, and V_{eK} follows.) In both cases the exact value lies strictly between the two constructions, with no adjustable quantity on either side: $D_{2\mathrm{hom}} < D_e < D_{2\mathrm{split}}$ and $E_{2\mathrm{split}} < E < E_{2\mathrm{hom}}$. This is the model's energy statement. An intermediate construction that reproduced the exact value would be a fit.

The geometry is sharper than the energy. The 2split H_2 converges in the axisymmetric reduction to $R_e = 1.368\,a_0$, short of the exact 1.401 by 2.3% and flat to 0.2% under a 2.5-fold mesh refinement.

5.3 What places the 2split side

The 2split energy is not an error in the solution: two independent solvers agree, the energy converges, and every term is counted. Its components show what the free wall does. Against exact, the nuclear attraction is larger (centroid of electron 1 at -0.785, the nucleus at -0.70) and V_{ee} smaller: a Neumann wall lets each electron keep full amplitude right up to the midplane—in the bond region, where both nuclei attract it—at no kinetic cost, and the other electron, confined strictly to the other side, never competes for that region. Charging the wall shows this directly. With a surface energy $\frac{\beta}{2}\int_{\partial\Omega_k} u_k^2$ on the interface:

β	T	V_{eK}	V_{ee}	centroid
0	1.291	-3.843	0.545	-0.785
0.15	1.277	-3.803	0.526	-0.832
0.5	1.265	-3.695	0.485	-0.936
2	1.281	-3.255	0.381	-1.269
10	1.238	-2.607	0.288	-1.713

As soon as the wall costs anything the density leaves the interface for the far side of its own nucleus and the nuclear attraction falls, with the kinetic energy almost unchanged. The free cut at the wall is the model's central assumption; it sets the 2split energy, while the 2hom pair has no wall.

5.4 Beyond two electrons

The same two constructions apply to any pair in a molecule. For LiH, with Li as a +1 pseudonucleus ($r_c = 1.99$ from its ionisation energy) and the bonding pair as 2split (Li valence and H electron in separate domains) or 2hom (one pair over Li^+ and a bare proton, Section 6.1), D_e is 0.063–0.073 (2split) and 0.095 (2hom) against the exact 0.092. For BH (pseudo-B +3 with a 2hom lone pair) D_e is 0.276 (2split) and 0.155 (2hom) against 0.135 (Table 1). In both, the 2hom bonding pair is the closer.

The picture also tells where each construction belongs. A pair that shares one region—a core $1s^2$, or a bonding pair near equilibrium—is naturally 2hom. Electrons that must avoid each other need 2split: the 2hom H^- is unbound, exactly as in Hartree–Fock, while the 2split H^- binds; and a 2hom pair cannot dissociate a bond into two neutral atoms. We report both, and choose by this rule.

6 Results

[In progress.] Bond lengths from an R-scan at three meshes ($h = 0.10, 0.07, 0.05$) per bond with `molecule.js`, extrapolated in h, with the kernel clamp pinned so that the ladder refines a fixed problem; axisymmetric values beside them for X_2 only. Energies for both constructions, 2hom and 2split, at the computed geometry.

		bond length (a_0)				binding energy D_e (E_h)		
molecule	construction	3D	axisym	exp.	error	2hom	2split	exp.
H_2	1+1	pending	1.368	1.401	−2.3%	0.138	0.281	0.1745
LiH	Li^+ (r_c) + pair	2.85	—	3.015	−5%	0.095	0.063–0.073	0.092
BH	B 2+3 (2_{hom}+1)	2.29	—	2.329	−1.6%	0.155	0.276	0.135
NH	N 2+2+3	pending	—					
OH	O +3 (2_{hom}+1)	1.79	—	1.832	−2%	0.241	0.371	0.170
BeH_2	Be $2+2_{split}$	2.44	—	2.52[a]	−3%	0.248[b]	n.c.	0.2231
HF	F 2+4+3	pending	—					
NaH	Na	pending	—					
H_2O	O +3 (1+1+1)	obs.[c]	—	1.809	—	0.651	n.c.	0.37
NH_3	N +3 / +5	obs.[c]	—	1.913	—	1.91 / 3.45	—	0.47
CH_4		pending	—					

Table 1: Bond lengths and the 2hom and 2split binding energies against experiment. H_2 energies are axisymmetric ($R = 1.4$, $h = 0.1$); LiH is three-dimensional (box $20\,a_0$, $h = 0.1$), its bond length from the 2hom construction (2split: 2.98–2.99). BH is three-dimensional (box $12\,a_0$, $h = 0.1$, pseudo-B $r_c = 1.08$), $R = 2.1, 2.3, 2.5$; bond length from 2hom (2split: 2.38). BeH_2: pseudo-Be $r_c = 1.37$, box $12\,a_0$, $R = 2.3, 2.5, 2.7$; [a]gas-phase r_0 (vibrationally averaged); [b]atomisation energy (two bonds), experiment $0.2231\,E_h$ [6]; n.c.: the 2split runs do not converge (the two sides trade cells and the symmetry breaks). OH: pseudo-O +3 (only the outer 3-shell carried, as 2hom lone pair +1 behind a plane), $r_c = 0.59$, $R = 1.65, 1.83, 2.0$; 2split $R_e \approx 1.69$; dipole 3.6 D (2hom), 2.7 D (2split) against 1.66. [c]Single point at the observed geometry; the binding is the atomisation energy. H_2O: pseudo-O +3, the outer shell as three single electrons in 120° wedges, $r_c = 0.65$; dipole +1.01 D against 1.85, right sign. NH_3: pseudo-N +3 (lone pair frozen, $r_c = 0.81$) and +5 (lone pair a 2hom domain, $r_c = 0.82$); see Section 7. The axisymmetric reduction is used for X_2 only. Remaining rows to be filled from the runs in progress.

6.1 LiH: pseudo-Li and one bonding pair

LiH is the simplest polar bond with an inner shell, and is viewed as ionic (Li^+H^-) rather than covalent: its dipole, 5.88 D, is 77% of that of point charges ±1 at $3.015\,a_0$ (7.66 D). The model reduction reflects this: the Li $1s^2$ core is excluded by a fixed sphere of radius $r_c = 1.99\,a_0$ around the Li nucleus, which then carries charge +1 (pseudo-Li); the two valence electrons form one 2hom pair over pseudo-Li and a bare proton. The radius is the one at which the pseudo-Li atom alone reproduces the Li ionisation energy; the one-dimensional all-electron Li atom places its own core–valence interface at 1.98–2.05 a_0 over boxes 6–20 a_0, within 3% of it. All runs are `molecule.js`, $h = 0.1$, box $20\,a_0$; binding energies are $E(\mathrm{Li}) + E(\mathrm{H}) - E(\mathrm{LiH})$ with both atoms built the same way and in the same box.

R	E	$D(R)$	μ (D)
2.6	−0.76776	0.0880	—
2.8	−0.77436	0.0946	6.45
3.0	−0.77259	0.0928	6.53
3.2	−0.76709	0.0873	6.66
3.4	−0.75999	0.0802	6.82
exp.		0.092 at 3.015	5.88

A fit through the five points gives $R_e = 2.85\,a_0$ (2.81–2.89 over three-, four- and five-point fits), 5%

short of experiment, and $D_e = 0.0947\,E_h$, 3% over. At $R = 3.0$ the dipole is 6.53 D, of the right sign (Li^+H^-) and 11% high; it is 86% of the point-charge value at that R, so the pair, free to distribute itself over both centres, settles largely on H. Nothing was adjusted to LiH. The 2split on the same pseudo-Li (Li valence and H electron in separate domains) gives a slightly longer bond, $R_e = 2.98$–2.99 (-1%), and under-binds, $D_e = 0.063$–0.073 (Section 5.4). The core is a hard exclusion with a radius set by the atom: a frozen-core model, stated as such. A 2hom core held inside the same fixed sphere is equivalent for the valence, since a spherical charge inside the sphere acts outside it as a point charge. The sphere must be fixed: a free boundary between core and valence carries no Pauli cost, and with it the valence presses into the core and $E(R)$ falls monotonically as H approaches.

6.2 BH and BeH_2

BH. Pseudo-B +3 carries its outer 2+3 shell as a 2hom lone pair and one electron, the two separated by a fixed surface through the B nucleus; the B|H boundary is free. Two divisions of the 3 were run, each with its own r_c fitted to the B ionisation energies (box $12\,a_0$, $h = 0.1$): a plane perpendicular to the bond (each domain half the solid angle, $r_c = 1.08$), and a cone about the bond holding one third of the solid angle for the single electron and two thirds for the pair (half-angle 70.5°, $r_c = 1.11$), in proportion to their electrons. For the bond, 2hom (one pair over the B front and a bare proton):

2hom BH	R_e	D_e	μ (D)
plane, $\frac{1}{2}$:$\frac{1}{2}$	2.29	0.155	3.5
cone, $\frac{1}{3}$:$\frac{2}{3}$	2.17	0.224	1.09
exp.	2.329	0.135	1.27

The proportional division corrects the dipole, by moving charge behind B, and costs bond length and binding: the pair confined to the cone draws the proton in. The plane gives bond and energy, the cone the dipole; neither gives all three. 2split binds more strongly in both (0.276 and about 0.34), with dipoles 4.0 and 1.2–1.3 D. Table 1 uses the plane.

BeH_2. Pseudo-Be +2 (2+2, $r_c = 1.37$) carries its two valence electrons as single electrons on either side of a fixed plane through Be, each facing one H. With each bond a 2hom pair over a Be half and a bare proton, $R_e = 2.44$ against the gas-phase $r_0 = 2.52$ (-3%), and the atomisation energy is $0.248\,E_h$ against 0.2231 [6] (+11%). The 2split construction does not settle: the two H domains and the two Be halves trade cells without end, the symmetry breaks (a residual dipole of 0.2 D), and the convergence window never closes.

7 Limitations

Polar bonds. Where the two sides of a bond differ strongly, the true domain boundary curves around the smaller or more weakly bound atom, so heteronuclear molecules are computed in three dimensions only. LiH is recovered once its valence electrons form one bonding pair over an excluded core (Section 6.1); with one electron per domain the dipole remains wrong in sign, so for polar bonds the pair construction, not the boundary shape, is decisive. NaH and HF are still to be run this way.

Inner shells. Pauli exclusion is included through disjoint domains: electrons in different domains do not overlap. A 2hom pair overlays two electrons of opposite spin in one domain, as Pauli permits,

and so captures that part of it too. In standard quantum mechanics exclusion is the antisymmetry of the many-electron wave function; imposed as orthogonality of, say, $2s$ to $1s$, it puts a node in the valence orbital inside the core, and the kinetic energy of that node keeps the valence out (the term a pseudopotential replaces). Disjoint domains are orthogonal by construction, $\int u_k u_l = 0$, but by separation rather than by a node: the valence amplitude is free at the wall and pays no node energy, and a free boundary between a shell and the valence, placed by continuity, carries no counterpart of that cost. Inner shells therefore cannot be represented by domains with free walls—the valence presses into them and the bond collapses—and are excluded by a fixed core radius per atom. This is a frozen-core model, and its radius is a property of the atom, set by its ionisation energy.

Strong pseudo-cores. The same missing cost limits the valence a pseudo-atom may carry. With cores of charge up to +3 (LiH, BeH_2, BH, OH, and H_2O with pseudo-O +3) the 2hom binding is within 0.003–0.3 E_h of experiment. With +5 and +6, carrying the whole valence shell, the valence floods the protons at no kinetic price: H_2O with pseudo-O +6 over-binds by 2.5 E_h, NH_3 with pseudo-N +5 by 3 E_h, the kinetic energy rising by less than a third of what the added H electrons alone should cost. Freezing the lone pair instead (pseudo-N +3) keeps the error to 1.4 E_h but leaves a spherical core and inverts the dipole of NH_3. The missing term is the Pauli kinetic energy between domains; until it is supplied, molecules whose heavy atom carries more than three valence electrons are outside the model's range.

Large molecules. The last reduction step (reduced valence charges, $h = 0.25$–$0.4\,a_0$) has been checked against reference chemistry only on small fragments: the formamide dimer and a five-residue polyglycine helix, both on 200^3, give bond and hydrogen-bond geometries in line with reference values. Folding runs on whole proteins use contact restraints; without them the fold is not found.

Contacts between molecules. A new domain boundary in a region of high density adds the same uncharged wall as the 2split bond, without a 2hom alternative. Intermolecular energies are outside the model's range, and so is dispersion.

Energies across compositions. Binding and reaction energies compare systems with different domain structures, whose errors differ; only differences within one composition (conformations, scans along a coordinate) are quotable.

8 Conclusion

Solved as it stands, the hard-partition model of RealQM carries every electron pair in one of two parameter-free constructions, 2split and 2hom, the limits of complete and of no avoidance; for H_2 and He the exact energy lies between them. A shared pair (core, bonding pair) is 2hom, and for LiH and BH the 2hom bond is the closer to experiment. Bond lengths are good where the domain boundaries are flat or fixed by symmetry. Inner shells must be excluded by a fixed core radius, since a free wall has no Pauli cost; with that, LiH as one 2hom bonding pair has its bond within 5%, its binding within 3% and a dipole of the right sign. Other polar bonds are [**pending**]. Nothing is adjusted to molecular data; each atom contributes one core radius, set from its ionisation energies.

References

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[3] K. P. Huber and G. Herzberg, *Molecular Spectra and Molecular Structure IV: Constants of Diatomic Molecules*, Van Nostrand Reinhold (1979).

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[5] Public code, the molecule simulator, and an interactive gallery: `https://claes542.github.io/RealMolecule/gallery.html` (source repository: `https://github.com/Claes542/RealMolecule`).

[6] BeH_2 atomisation energy, $0.2231\,E_h$; U.S. DOE Office of Scientific and Technical Information, `osti.gov`.